

LEPHALALE, South Africa

WMO: 681550

Lat: 23.6767S

Lon: 27.7050E

Elev: 839

StdP: 91.64

Time Zone: 2.00 (E02)

Period: 98-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	4.5	5.9	-12.9	1.4	21.2	-9.2	1.9	18.7	5.4	18.9	4.8	19.3	0.9	190	0.291

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	10.7	37.3	18.9	35.9	18.9	34.6	19.1	23.9	29.9	23.2	29.1	22.6	28.3	2.2	0

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
22.5	19.1	26.0	21.7	18.2	25.6	21.0	17.4	25.2	76.6	29.7	74.0	29.2	71.2	28.4	26.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 5.0	4.3	3.8	DB	1.8	40.1	1.4	1.9	0.8	41.5	0.0	42.6	-0.8	43.7	-1.8	45.1	(4)
(5)			WB	-1.3	24.3	0.9	1.2	-1.9	25.1	-2.5	25.8	-2.9	26.4	-3.6	27.3	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	22.3	26.3	26.6	25.0	21.9	18.8	15.9	15.5	18.6	22.6	24.9	25.8	26.2	(6)	
	DBStd	4.76	2.30	2.29	2.43	2.39	2.56	1.98	1.90	2.80	2.96	3.36	3.12	2.88	(7)	
	HDD10.0	0	0	0	0	0	0	0	0	0	0	0	0	0	(8)	
	HDD18.3	229	0	0	1	2	24	78	89	31	3	1	1	0	(9)	
	CDD10.0	4497	507	464	465	357	274	176	171	266	379	462	474	503	(10)	
	CDD18.3	1683	248	231	207	108	39	3	2	39	132	206	224	244	(11)	
	CDH23.3	16873	2360	2212	1748	807	434	135	123	549	1445	2261	2348	2451	(12)	
(13)	CDH26.7	7434	1026	998	739	273	102	14	10	173	679	1130	1146	1143	(13)	
(14)	Wind	WSAvg	1.6	1.6	1.5	1.4	1.3	1.2	1.2	1.4	1.6	2.0	2.1	1.9	1.7	(14)
(15) Precipitation	PrecAvg	403	76	58	51	27	6	5	1	2	4	32	58	84	(15)	
	PrecMax	604	146	109	195	85	39	66	12	13	28	83	165	167	(16)	
	PrecMin	220	17	17	4	0	0	0	0	0	0	1	1	34	(17)	
	PrecStd	102	41	30	47	25	9	14	3	3	6	22	37	35	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	37.9	37.9	36.2	34.3	31.9	28.2	28.0	32.8	36.5	39.3	39.1	38.4	(19)	
		MCWB	19.9	20.4	18.8	18.6	15.9	13.5	12.2	14.7	16.2	17.0	18.0	20.9	(20)	
	2%	DB	35.1	35.4	34.2	31.9	29.2	26.4	26.2	30.7	34.6	36.7	36.5	36.0	(21)	
		MCWB	20.8	20.4	19.2	18.3	15.6	13.2	12.3	14.4	15.7	17.1	18.7	20.5	(22)	
	5%	DB	33.3	33.8	32.7	29.9	27.5	24.9	24.7	28.7	32.9	34.7	34.4	34.2	(23)	
		MCWB	20.6	20.6	19.2	18.2	15.2	12.9	12.7	13.8	15.1	17.3	18.8	20.7	(24)	
	10%	DB	31.8	32.1	30.8	28.0	25.8	23.1	23.0	26.5	30.9	32.6	32.7	32.4	(25)	
		MCWB	20.7	20.2	19.4	18.1	15.0	12.8	12.4	13.4	15.0	17.1	18.8	20.5	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	25.2	24.1	23.8	23.8	21.2	16.5	15.6	18.1	19.1	20.7	23.4	24.6	(27)	
		MCDB	29.8	31.3	28.2	26.5	26.0	23.5	22.6	25.7	30.2	30.2	31.2	31.6	(28)	
	2%	WB	23.8	23.3	22.7	22.1	19.0	15.2	14.6	16.5	17.9	20.0	22.2	23.7	(29)	
		MCDB	29.5	29.8	27.1	26.0	22.6	22.6	22.0	25.9	29.3	29.2	28.8	30.0	(30)	
	5%	WB	23.0	22.7	22.0	20.6	17.1	14.5	13.9	15.6	17.2	19.2	21.2	22.9	(31)	
		MCDB	29.5	28.6	27.0	25.8	24.3	21.7	21.5	25.2	27.9	28.5	28.1	28.6	(32)	
	10%	WB	22.2	22.0	21.2	19.5	16.1	13.7	13.1	14.6	16.4	18.6	20.6	22.1	(33)	
		MCDB	28.1	27.5	25.9	25.2	23.5	20.7	20.8	24.3	26.8	27.9	27.8	28.1	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	10.2	10.7	10.9	12.2	15.1	16.1	16.1	16.2	15.4	13.3	12.0	10.7	(35)	
		MCDBR	13.1	13.5	14.3	15.4	17.0	18.6	18.7	18.8	18.5	16.5	14.8	13.6	(36)	
	5% WB	MCWBR	3.5	3.6	3.8	4.9	6.2	8.1	8.6	7.5	6.6	5.0	3.8	3.4	(37)	
		MCWBR	10.5	10.0	9.7	11.0	13.1	14.5	15.0	15.3	14.8	12.8	11.9	10.9	(38)	
(40) Clear-Sky Solar Irradiance	taub		0.394	0.396	0.385	0.355	0.314	0.300	0.296	0.354	0.409	0.429	0.394	0.394	(40)	
		taud	2.376	2.369	2.382	2.430	2.503	2.513	2.516	2.341	2.207	2.201	2.344	2.376	(41)	
	Ebn at Noon		950	934	916	902	903	897	915	888	879	893	944	951	(42)	
		Edh at Noon	131	129	122	107	91	86	88	115	143	151	134	131	(43)	
(44) All-Sky Solar Radiation	RadAvg		6.75	6.35	5.72	4.93	4.60	4.10	4.42	5.23	6.05	6.54	6.67	6.63	(44)	
	RadStd		0.52	0.63	0.57	0.43	0.19	0.24	0.12	0.23	0.32	0.40	0.62	0.44	(45)	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Regional (2 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page